

1. Using the Maxwell-Boltzmann speed distribution, find the following quantities.

Given the Maxwell-Boltzmann speed distribution:

$$f(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-mv^2/2k_B T}$$

(a) $\langle v \rangle = \int_0^\infty v f(v) dv$

$$\begin{aligned} \langle v \rangle &= \int_0^\infty v f(v) dv \\ &= \int_0^\infty 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^3 e^{-mv^2/2k_B T} dv \\ &= 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} \int_0^\infty v^3 e^{-mv^2/2k_B T} dv \end{aligned}$$

Using the known Gaussian integral

$$\begin{aligned} \int_0^\infty x^3 e^{-\alpha x^2} dx &= \frac{1}{2\alpha^2}, \\ \langle v \rangle &= 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} \frac{1}{2(m/2k_B T)^2} \\ &= 2\pi^{-1/2} \left(\frac{m}{2k_B T} \right)^{-1/2} \\ &= \sqrt{\frac{8k_B T}{\pi m}}. \end{aligned}$$

$$\langle v \rangle = \sqrt{\frac{8k_B T}{\pi m}}$$

(b) $\langle v^2 \rangle = \int_0^\infty v^2 f(v) dv$

$$\begin{aligned} \langle v^2 \rangle &= \int_0^\infty v^2 f(v) dv \\ &= \int_0^\infty 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^4 e^{-mv^2/2k_B T} dv \\ &= 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} \int_0^\infty v^4 e^{-mv^2/2k_B T} dv \end{aligned}$$

Using the known Gaussian integral

$$\begin{aligned}
 \int_0^\infty x^4 e^{-\alpha x^2} dx &= \frac{3}{8} \sqrt{\frac{\pi}{\alpha^5}} \\
 \langle v^2 \rangle &= 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} \cdot \frac{3}{8} \sqrt{\frac{\pi}{(m/2k_B T)^5}} \\
 &= \frac{3}{2} \left(\frac{m}{2k_B T} \right)^{3/2} \left(\frac{m}{2k_B T} \right)^{-5/2} \\
 &= \frac{3k_B T}{m}.
 \end{aligned}$$

$$\boxed{\langle v^2 \rangle = \frac{3k_B T}{m}}$$

(c) $v_{rms} = \sqrt{\langle v^2 \rangle}$

$$\begin{aligned}
 v_{rms} &= \sqrt{\langle v^2 \rangle} \\
 &= \sqrt{\frac{3k_B T}{m}}.
 \end{aligned}$$

$$\boxed{v_{rms} = \sqrt{\frac{3k_B T}{m}}}$$

(d) $\langle E_{KE} \rangle$

$$\begin{aligned}
 \langle E_{KE} \rangle &= \frac{1}{2} m \langle v^2 \rangle \\
 &= \frac{1}{2} m \frac{3k_B T}{m} \\
 &= \frac{3k_B T}{2}.
 \end{aligned}$$

$$\boxed{\langle E_{KE} \rangle = \frac{3k_B T}{2}}$$

2. Show that

$$\bar{E} = -\frac{1}{Z} \frac{\partial Z}{\partial \beta} = -\frac{\partial \ln Z}{\partial \beta}.$$

The partition function is defined as $Z = \sum_s e^{-\beta E_s}$ where $\beta = \frac{1}{k_B T}$.

$$\begin{aligned}
 \frac{\partial Z}{\partial \beta} &= \sum_s -E_s e^{-\beta E_s} \\
 \bar{E} &= -\frac{1}{Z} \sum_s -E_s e^{-\beta E_s}
 \end{aligned}$$

$$\begin{aligned}
&= \frac{\sum_s E_s e^{-\beta E_s}}{Z} \\
-\frac{\partial \ln Z}{\partial \beta} &= -\frac{1}{Z} \frac{\partial Z}{\partial \beta} \\
&= \frac{\sum_s E_s e^{-\beta E_s}}{Z}.
\end{aligned}$$

$$\boxed{-\frac{1}{Z} \frac{\partial Z}{\partial \beta} = -\frac{\partial \ln Z}{\partial \beta}}$$